[NOT MEASUREMENT SENSITIVE]
A-A-50425
June 26, 1996

COMMERCIAL ITEM DESCRIPTION

CLEANING COMPOUND, HALOGENATED HYDROCARBON, FOR OXYGEN SYSTEM COMPONENTS, AEROSOL (SPOT CLEANER)

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) covers one type of spot cleaner, packaged in a pressurized container with a non-ozone depleting substance propellant, for oxygen system component cleaning.

2. SALIENT CHARACTERISTICS.

2.1 Material. The cleaning compound shall be a mixture of 1,1-dichloro-1-fluoroethane (hydrochlorofluorocarbon, HCFC-141b) (Chemical Abstract Service [CAS] Registry Number 1717-00-6), and isopropyl alcohol (CAS 67-63-0) with carbon dioxide (CAS 124-38-9) as the propellant.

2.2 Composition. The wipe cleaner shall be a nonflammable, evaporating, non-residue solvent system as specified in table I.

2.2.1 Material safety data sheet. A material safety data sheet shall be prepared and submitted in accordance with FED-STD-313, meeting the requirements of Code of Federal Regulation 29 CFR 1910.1200. Questions pertinent to the effect of the cleaner on the health of personnel when used for its intended purpose shall be referred by the acquiring activity to the appropriate medical service who shall act as adviser to the acquiring activity.

Beneficial comments, recommendations, additions, deletions, clarifications and any other data which may improve this document should be sent to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-3 Highway 547, Lakehurst, NJ 08733-5100.

FSC 6850

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TABLE I. Composition and solvent purity.

Component	Component purity % by wt	Composition % by wt
1,1-dichloro-1-fluoroethane (HCFC-141b)	99.50 1/	90-94
Isopropyl alcohol (Isopropanol)	2/	3-5
Carbon dioxide propellant	3/	3-5

1/ Component purity shall be tested as specified in ASTM-D3447 if using bulk HCFC-141b material other than A-A-50427.
2/ Anhydrous, chemically pure.
3/ Sufficiently pure to meet requirements of this commercial item description.

2.3 Product characteristics.

2.3.1 Appearance. The solvent mixture shall be a clear, colorless liquid (water white) with no evidence of particles. The appearance shall be determined by visually examining 500 mL of solvent for clarity, color and solids in a clear container against a white and black background. The inspection shall be performed by a person with normal visual acuity, natural or corrected, under bright light of not less than 100 foot-candles. A general purpose flashlight with 2 fresh batteries and a fresh lamp, positioned 18 inches perpendicular to a surface will illuminate that surface with an intensity of not less than 100 foot-candles.

2.3.2 Moisture content. There shall be no more than 50 parts per million (ppm) by weight moisture, when tested as specified in ASTM-D3401.

2.3.3 Chloride ion. There shall be no more than 1 ppm chloride ion present in the product when tested in accordance with ASTM-D3443.

2.3.4 Nonvolatile residue. There shall be less than 20 ppm by weight nonvolatile residue when tested as specified in ASTM-D2109 or ASTM-F331. The sample for test shall be collected under clean room conditions immediately after filling. Unless otherwise specified in the contract or order, the nonvolatile residue shall be determined, during production, by sampling the first, last, and every 1250th pressurized containers manufactured at one plant, from the same materials under essentially the same conditions provided the operation does not exceed 24 hours.

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2.4 Pressurized container. The pressurized container shall include a valve designed to permit the cleaner to be sprayed in an upright position. The activating device shall deliver the cleaner in a conical spray and shall be so designed to accommodate the attachment of a plastic or metal extension tube, which is also supplied. The container shall have a snap cover attachment to protect the valve when not in use.

2.4.1 Leakage. The pressurized container shall not leak nor become distorted after the following procedure:

> The solvent shall be sprayed for three seconds followed by a two second pause. This operation shall be repeated for a total of five spray/no spray cycles. The can shall then be completely submerged for 5 minutes in a saturated sodium chloride or brine solution maintained at 54° ±1° C (130° ±2° F). During this time, the emission of bubbles shall be noted. The can shall be removed from the solution and observed for distortion, then re-submerged in the solution for an additional three minutes to check for any leakage.

2.4.2 Weight. The weight of the cleaner shall be not less than 15 ounces when determined by weighing a full container to the nearest 0.1 ounce. The contents shall be sprayed for three minutes, followed by a one minute pause. This cycle shall be repeated until there is no further evidence of spray. The can shall then be punctured to completely empty any residual contents. Reweigh the container to the nearest 0.1 ounce.

2.5 Shelf life. The cleaning compound shall meet all the requirements of this CID after storage in unopened containers for not less than one year.

2.6 Warning labels. Warning labels shall include the following information. Because of size constraints, the entire label may be too large for the available area. In those instances, it is recommended that the acquiring activity approve a supplier condensed label containing as much information as possible.

DICHLOROFLUOROETHANE
HCFC-141b

WARNING

> Eye contact with vapor causes irritation. Contact with liquid can cause chemical burns. Prolonged/repeated skin contact may cause dryness/irritation. Exposure can also result in freezing of skin tissue (frost bite).

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Over-exposure to vapor may cause respiratory system irritation, headache, dizziness and central nervous system (CNS) depression.
Exposure to higher levels may also result in increasing sensitivity of the heart to adrenaline. Cardiac arrhythmia could result.
Exposure to very high levels could result in death due to dilution/displacement of available oxygen, resulting in death by asphyxiation. Low areas are especially susceptible to oxygen displacement. Taste /odor do not provide adequate warning of high vapor levels.
Do not use in the general vicinity of welding, open flames or hot surfaces. This could result in the decomposition of this material to irritating and toxic gases and carbonyl halides.

In case of spills/significant release to work space, warn others and leave space immediately.

Use smallest quantity possible to properly perform the operation.

Use only with adequate ventilation . Refer to material safety data sheet for additional information and required precautions for use. Consult medical department authority for guidance in control and proper use of this material, and specific requirements and methods for monitoring vapor levels in the workspace.

Have an emergency escape breathing device (EEBD) immediately available for any operation where overexposure to product could occur unless immediate escape to a vapor-free area is possible. Be sure you are trained in the proper care, inspection and use of this device.

Ensure storage areas which contain this product are properly ventilated prior to entry.

FIRST AID

Inhalation: Immediately remove person to fresh air. If breathing has stopped, give artificial respiration. In cases of excessive exposure, pay particular attention to the potential for cardiac arrhythmia. Oxygen may be used in treating such cases if administered by medical department personnel. Use of vaso-pressor drugs, for example epinephrine (adrenaline), is inadvisable because these drugs may induce cardiac arrhythmia. If high temperature decomposition products may have been inhaled, observe the patient for immediate or delayed signs of pulmonary edema. In any of these cases, obtain prompt medical attention.

NOTE TO PHYSICIANS: Overexposure to dichlorofluoroethane has caused cardiac sensitization to epinephrine. Cardiac arrhythmia, including

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ventricular fibrillation could occur if epinephrine or one of its congeners is administered to patients exposed to high concentrations of dichlorofluoroethane. Medical use of epinephrine or any of its congeners is contra-indicated except for patients with no arterial profusion.

Skin contact: If pain or freezing of tissue occurs, seek prompt medical attention. If no freezing of the skin is apparent, wash areas with warm water. If material soaks through clothing, remove clothing and wash skin with soap or mild detergent and water. If irritation persists after washing, obtain medical attention. Do not re-use contaminated clothing until it is thoroughly cleaned to remove the material.

Eye contact: Immediately wash with large amounts of potable water for a minimum of 15 minutes. Avoid use of any other agents such as oil or saline solutions in the eye. Use of any other agent or dressing should only be administered by the cognizant medical department representative.

Contact lens should not be worn when using this chemical.

Ingestion: Do not induce vomiting. Seek medical attention.

3. REGULATORY REQUIREMENTS.

3.1 Environmental considerations. The product delivered under this CID shall be in accordance with the environmental requirements of Code of Federal Regulation 40CFR.

3.2 Transportation. The Transportation of the product delivered under this CID shall be in accordance with Code of Federal Regulations 49 CFR 171-190.

3.3 Pressurized container. The pressure of the container shall be within DOT requirements.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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4.2 Market acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this commercial item description:

a. The company furnishing the spot cleaner must have been producing a product meeting the requirements of this commercial item description for at least one year.

b. The company must have offered the spot cleaner to a government laboratory for inspection.

5. PACKAGING.

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 Copies of the Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, US Government Printing Office, Washington DC 20402.

6.1.2 Federal Government activities may obtain copies of Federal Standard No. 313 from established distribution points within their agencies. Other activities may obtain the standard for bidding purposes from the General Services Administration, Federal Supply Service Bureau, Specification Section, Suite 8100, 479 L'Enfant Plaza, SW, Washington, DC 20407.

6.1.3 The following test methods are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19103:

ASTM-D2109	Test Method for Nonvolatile Matter in Halogenated Organic Solvents and their Admixtures
ASTM-D3401	Test Method for Water in Halogenated Organic Solvents and their Admixtures
ASTM-D3443	Test Method for Chloride in Trichlorotrifluoroethane,
ASTM-D3447	Test Method for Purity of Trichlorotrifluoroethane
ASTM-F331	Test Method for Nonvolatile Residue of Halogenated Solvent Extract from Aerospace Components (Using Rotary Flash Evaporation)

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6.2 Government laboratory. Manufacturers wanting to have their cleaning compound solvent inspected may do so by contacting the Naval Air Warfare Center Aircraft Division, Mail Stop 3, Bldg 2188, Code 4.3.4.1, 48066 Shaw Road, Patuxent River, MD 20670-5304.

6.3 Use. The use of this compound for cleaning oxygen system components should be accomplished only with approved procedures.

MILITARY INTERESTS:

Custodians:
Army - EA
Navy -AS
Air Force - 68

Preparing Activity:

Navy - AS
DoD Project 6850-1167

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